CHAPTER 1

1.1 INTRODUCTION

The CNC laser engraver machine is a platform to precisely control motion. This can easily be a laser engraver or a laser engraver CNC machine.

CNC stands for Computer Numerical Control and has been around since the early 1970's. Prior to this, it was called NC, for Numerical Control. (In the early 1970's computers were introduced to these controls, hence the name change.) While people in most walks of life have never heard of this term, CNC has touched almost every form of manufacturing process in one way or another. If you'll be working in manufacturing, it’s likely that you'll be dealing with CNC on a regular basis.

In a typical NC system, the motion and machining instructions and the related numerical data, together called a part program, used to be written on a punched tape. The part program is arranged in the form of blocks of information, each related to a particular operation in a sequence of operations needed for producing a mechanical component. The punched tape used to be read one block at a time. Each block contained, in a particular syntax, information needed for processing a particular machining instruction such as, the segment length, its cutting speed, feed, etc. These pieces of information were related to the final dimensions of the work piece (length, width, and radii of circles) and the contour forms (linear, circular, or other) as per the drawing. Based on these dimensions, motion commands were given separately for each axis of motion. Other instructions and related machining parameters, such as cutting speed, feed rate, as well as auxiliary functions related to part clamping, are also provided in part programs depending on manufacturing specifications such as tolerance and surface finish. Punched tapes are mostly obsolete now, being replaced by magnetic disks and optical disks Computer Numerically Controlled (CNC) machine tools, the modern versions of NC machines have an embedded system involving several microprocessors and related electronics as the Machine Control Unit (MCU). Initially, these were developed in the seventies in the US and Japan. However, they became much more popular in Japan than in the US. In CNC systems multiple microprocessors and programmable logic controllers work in parallel for simultaneous servo position and velocity control of several axes of a machine for contour cutting as well as monitoring of the cutting process and the machine tool. Thus, milling and boring machines can be fused into versatile machining centres. Similarly, turning centres can realize a fusion of various types of lathes. Over a period of time, several additional features were introduced, leading to increased machine utilization and reduced operator intervention.

1.2 SCOPE OF THE CAPSTONE PROJECT

Problem Statement: Our goal is to design and develop a CNC (Computer Numerical Control) laser engraver machine that can efficiently and accurately engrave various materials, such as wood, acrylic, metal, and leather. The machine should be capable of precise positioning and high-speed movement to achieve detailed and intricate designs.

Objectives: Design and develop a CNC Laser Engraver Project aims to provide a cutting-edge, reliable, and user-friendly solution for businesses, hobbyists, and artists, enabling them to achieve precise and efficient engraving on various materials.

Capstone Project Description: 1. Accuracy.

2. Repeatability.

3. Efficiency.

Capstone Project Deliverables:

1. **Machine Working:** The machine that can provides stability, precision, and durability for the machine's operation.

2. **User manual:** A guide for users to understand how to load, configure and use the machine.
3. **Source code:** The complete source code for the machine.
4. **Test results:** Results of the tests performed to check the accuracy and performance of the machine.

Key Milestones:

1. Project Conception and Initiation
2. Planning and Requirements
3. Project Design
4. Project Development
5. Testing

Constraints:

- **Technical constraints:** This could include the hardware and software requirements for the system, such as the controller, power supply, and lathe machine. The system should also be designed to work under different environmental conditions such as high humidity, and extreme temperatures.
- **Time constraints:** The project should be completed within a specific timeframe, with milestones set along the way to ensure that progress is being made.

Estimated Capstone project Duration: 12 weeks.

Estimated Capstone project cost: Rs. 30,540/-

CHAPTER 2

2.1 CAPSTONE PROJECT PLANNING

2.1.1 WORK BREAKDOWN STRUCTURE-DELIVERABLES

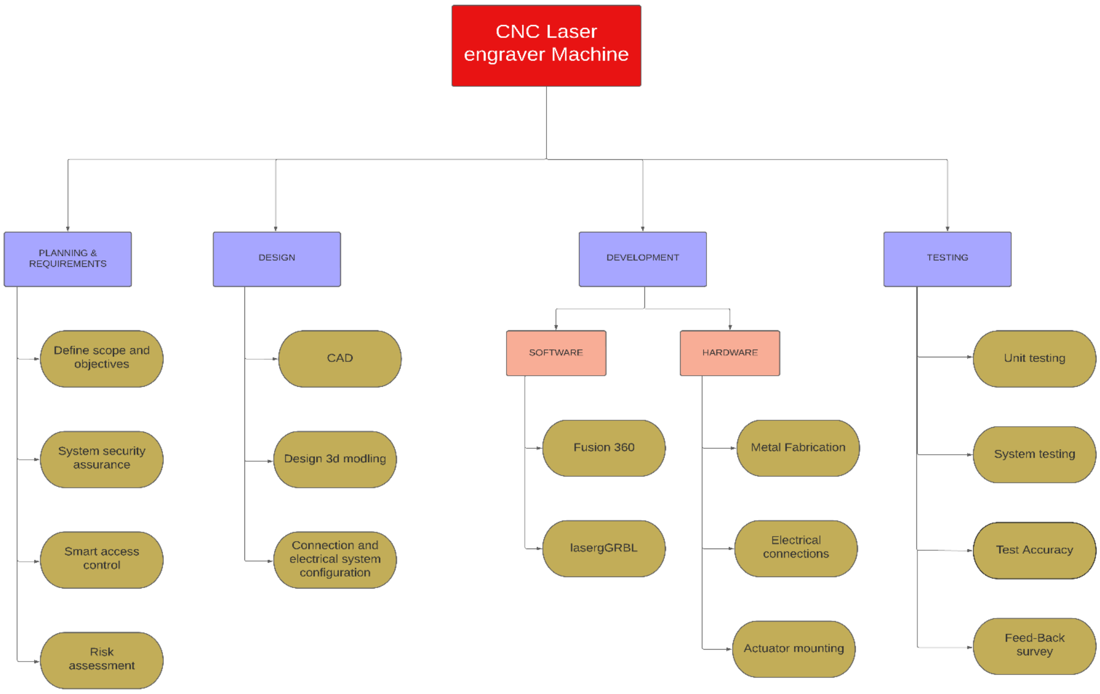


Figure no: 1

2.1.2 TIMELINE DEVELOPMENT – SCHEDULE

Task	Task Name	Start	Finish	Duration
Phase 1	**PRE-REQUIREMENTS TO START**	**27/2/2023**	**03/3/2023**	**04 Days**
Task 1	Research	27/2/2023	01/3/2023	03 Days
Task 2	Project Initiation	01/3/2023	03/3/2023	03 Days
Phase 2	**PLANNING & REQUIREMENTS**	**04/3/2023**	**17/3/2023**	**12 Days**
Task 1	Define scope and objectives	04/3/2023	11/3/2023	07 Days
Task 2	Smart access control and risk assessment	13/3/2023	17/3/2023	05 Days
Phase 3	**DESIGN**	**18/3/2023**	**29/3/2023**	**10 Days**
Task 1	CAD Design, Design 3d modelling and design the electrical system configuration	18/3/2023	29/3/2023	10 Days
Phase 4	**DEVELOPMENT**	**30/3/2023**	**16/5/2023**	**40 Days**
Task 1	Software Installation	30/3/2023	1/4/2023	25 Days
Sub task 1	Fusion 360	30/3/2023	11/4/2023	10 Days
Sub task 2	LaserGRBL	12/4/2023	29/4/2023	15 Days
Task 2	Metal Fabrication , Electrical connections and Actuator mounting	29/4/2023	16/5/2023	15 Days
Phase 5	**TESTING**	**16/5/2023**	**30/5/2023**	**14 Days**
Task 1	Unit testing and system testing	16/5/2023	23/5/2023	07 days
Task 2	Test documentation and feed-back survey	24/5/2023	01/6/2023	07 Days

Table no: 1

2.1.3 COST BREAKDOWN STRUCTURE (CBS)

SL No.	PARTICURALS	Qty.	COST IN RUPEES
1.	Nema 17	4	2,700/-
2.	Belt GT2 10mm	4	799/-
3.	Pulley 10mm GT2	4	499/-
4.	Idler pulley 10mm	8	2,392/-
5.	Power Supply	1	570/-
6.	Wire kit	1	400/-
7.	Bearings	45	1,245/-
8.	Arduino UNO	1	1,172/-
9.	CNC shield	2	240/-
10.	Motor driver	4	599/-
11.	Mounting hardware	-	500/-
12.	Filament	2kg	1,598/-
13.	5.5W Laser	1	17,826/-
TOTAL			**30,540/-**

Table no: 2

2.1.4 CAPSTONE PROJECT RISKS ASSESSMENT

- **Technical Risks:**
  - Malfunctioning of sensors.
  - Mechanical Failure.
  - Software Malfunction.
  - Electrical or Electronic Failure.
  - Calibration Issues.

- **Schedule Risks:**
  - Equipment delivery delays.
  - Resource availability.
  - Design changes.
  - Testing and calibration.
  - Unforeseen events.

- **Security Risks:**
  - Physical security.
  - Operator safety.
  - Intellectual property theft.
  - Compliance.

- **Legal Risks:**
  - Patent infringement.
  - Product liability.
  - Employment law.
  - Environmental regulations.
  - Contract disputes.

2.2 REQUIREMENTS SPECIFICATION

2.2.1 FUNCTIONAL

- **Nema 17 Stepper Motor:** Nema 23 stepper motors, a stepper motor, also known as step motor or stepping motor, is a brushless DC electric motor that divides a full rotation into a number of equal steps.

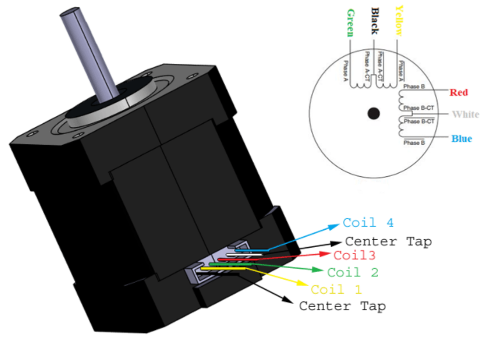


Figure No: 1 Figure No: 2

- **GT2 Belt 10mm:** 10mm GT2 Timing Belt is designed to convert rotational power from various motors such as stepper motors and servo motors into linear motion.

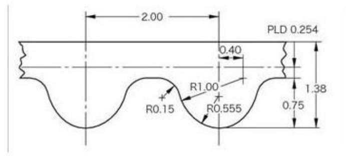


Figure No: 3 Figure No: 4

- **Pulley GT2**

  Aluminium 20T GT2 timing pulley has a high-quality shiny surface appearance and robust construction, longer working life. The rounded teeth of this tooth and belt are engaged together longer providing a smoother transition between teeth, good performance for Laser engraver.

Figure No: 5

- **POWER SUPPLY**

A switched-mode power supply, sometimes known as a switch-mode power supply or 'SMPS', is an electronic power supply that integrates a switching regulator for efficient electrical power conversion. Like other supplies, an SMPS transfers power from a DC or AC source to DC loads while converting voltage and current.

Figure No: 6

- **CONTROL BOARD:**

In controller we have used a Microcontroller called Arduion UNO board and two CNC shields and some stepper motor driver.

- **Arduion UNO:**

The Arduino Uno is an open-source microcontroller board based on the Microchip ATmega328P microcontroller (MCU) and developed by Arduino.cc and initially released in 2010.

Figure No: 7

- **CNC SHIELD**

The Arduino CNC Shield makes it easy to get your CNC projects up and running in a few hours.

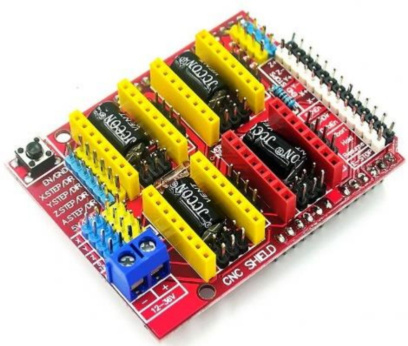


Figure No: 8

- **A4988 Stepper Motor Driver**

A4988 Stepper Motor Driver is a carrier board or breakout board for Allegro's A4988 DMOS Microstepping Driver with current limiting and overcurrent protection.

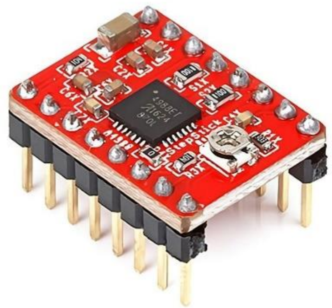


Figure No: 9

- **LASER MODULE**

It has a variable lenses to set the focus. The aluminium housing and forced air cooling ensures efficient cooling for optimal operation and long service life. The intensity of the laser can be adjusted via the TTL/PWM signal. Please see the table below for the materials that can be processed with this laser.

Figure No: 10

2.2.2 USER INPUT

The user input in a CNC laser engraving machine varies according on the machine and its interface. However, here are some examples of frequent sorts of user inputs:

- **Design File:** The user can supply a digital design file in a format that the CNC laser engraver machine can read. This might be a vector file (SVG, DXF) or a raster image file (BMP, JPEG) with the artwork or text to be engraved.
- **Settings:** The user can choose numerous engraving process settings. This may include laser power, engraving speed, engraving depth or intensity, scanning pattern, and other machine-specific variables.
- **Material Selection:** The user may be required to pick the type of material on which to engrave. Varied materials have varied qualities, therefore the machine may need different settings for each material to provide the best results.
- **Positioning and Alignment:** To properly position the laser head on the work piece, the operator may need to enter coordinates or utilise manual controls. This guarantees that the engraving is correctly and precisely set.
- **Start and Stop Commands:** After defining the design file, parameters, material, and placement, the user may begin the engraving process by issuing a start command. They can also use a stop command to pause or terminate the engraving operation.

It's worth noting that the user interface and input techniques varies amongst CNC laser engraving machines. Some machines may feature physical buttons, knobs, and touchscreens, whilst others may have software interfaces that allow users to enter data using a computer or a linked device.

2.2.3 TECHNICAL CONSTRAINTS

The technical constrains involved in the project CNC Laser engraver machine are:

- **Laser Power:** The engraving speed and depth are determined by the laser power. The laser's power output must be carefully selected to guarantee that it is sufficient for the required engraving results. Inadequate power may result in sluggish or shallow engraving, whilst high power may cause material damage or excessive heat.
- **Laser Type:** Different types of lasers, such as CO2 lasers and fibre lasers, are utilised in laser engravers. Each variety has unique properties such as power consumption, beam quality, wavelength, and maintenance requirements. The laser type used is determined by the materials to be engraved and the desired engraving quality.
- **Material Compatibility:** Laser engraving works well on particular materials such as wood, acrylic, glass, and certain metals. However, not all materials can be etched efficiently using a laser. The equipment must be constructed to accommodate the unique engraving materials, and the laser settings must be optimised accordingly.
- **Workspace Dimensions:** The maximum dimensions of the objects that may be engraved are determined by the size of the machine's working area. The machine's design must take into account the needed workspace size, including the X, Y, and Z axes, to handle varying work piece sizes.
- **Precision and Accuracy:** To generate intricate and crisp engravings, laser engraving demands great precision and accuracy. To achieve correct placement and movement, the machine's mechanical components, such as the linear guides, stepper motors, and control system, must be properly chosen and calibrated.
- **Cooling and ventilation:** Because laser engraving creates heat, suitable cooling and ventilation systems are required to prevent overheating. To ensure stable operation and preserve sensitive components, effective

cooling systems, such as water or air cooling, must be integrated into the machine design.

- **Control System and Software:** The CNC control system and engraving software are critical to the machine's performance. The laser strength, speed, and movement should all be precisely controlled by the control system. The programme should have simple user interfaces, handle several file formats, and provide complex functionality like as vector engraving and picture processing.
- **Safety Features:** Laser engravers produce intense laser beams that can be dangerous to human users. To prevent operators from unintentional laser exposure, the machine design must contain suitable safety features such as enclosure interlocks, emergency stop buttons, and laser safety eyewear.
- **Maintenance and serviceability:** The machine should be built to be simple to maintain and service. When troubleshooting, repairs, and component replacements are required, accessible components, modular designs, and comprehensive documentation can help.

2.3 DESIGN SPECIFICATION

2.3.1 SYSTEM DESIGN

2.3.1.1 SPECIFICATION:

Not easy to make a specs page when it can have most any specs you want. The idea behind this is to be extremely adaptable and inexpensive. Why buy a single use machinewhen you can have one that does it all for less than the cost of any single function machine. No proprietary hardware or software, build it any size and shape you want (more on this later).

2.3.1.2 SIZE:

All axis can be any length you prefer, anything over 3 ½′ (1M) would require small mid-span supports to increase rigidity, of course smaller is better. The kit comes with enough belt for up to 48″ of total outer X and Y axis dimensions (e.g.: 24″x24″, 36″x12″ or any other combination).

The smaller you make this the faster you can move it and the more rigid/accurateit will be, and more importantly the easier it will be to get the desired accuracy.

2.3.1.3 POWER:

X and Y axis are powered by 2 stepper motors each axis. The standard would be NEMA 17 in any torque preferably 50 OZ/in and above. No need for NEMA 23's or their required larger drivers, torque is not one of this machines issues.

Belts are used for their accuracy and price. Small belts are used because each axis has two of them. Ball screws are expensive, require tuning, and periodic maintenance / adjustments.

2.3.1.4 CONTROL:

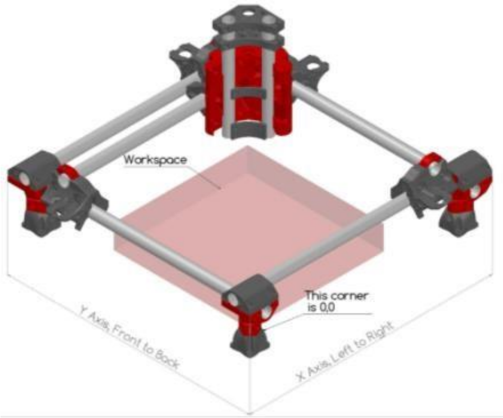
This is all controlled by any control board you like. The UltiMachine boards are recommended for the improved design and robust safety features. You can still use the common Ramps 1.4 or any others boards as well. There are other Marlin based boards, GRBL, and regular CNC boards as well as.

2.3.1.5 SOFTWARE:

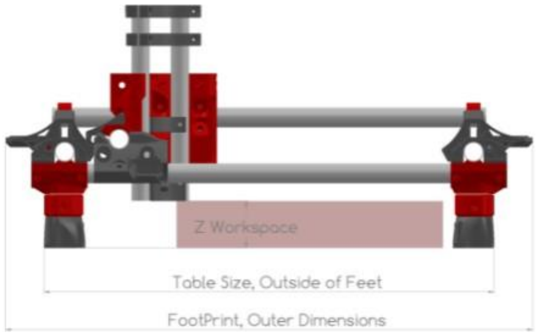
There are plenty of free or really inexpensive software options available for 3D printing, CAM, and CAD. I suggest Fusion 360, Repetier-host, ESTLCam, bCNC, etc.

LaserGRBL to run and control CNC Laser engraver machine.

2.3.1.6 SIZE CALCULATION:



Figures no: 11



Figures no: 12

UNITS are in mm

2.3.1.7 WORKSPACE

- X-Axis = 700mm
- Y-Axis = 700mm
- Z-Axis = 50 mm

2.3.1.8 TABLE SIZE

Length (mm)	Name
970	x table size (outer edges of feet)
979	y table size (outer edges of feet)

Table No: 4

2.3.1.9 TUBE LENGTHS

Length (mm)	Qty	Name
1004	2	x rails, sides
949	1	x rail, gantry
1013	2	y rails, sides
958	1	y rail, gantry
340	2	z rails
129	4	legs
7173	**total needed**	total tube length assuming 3mm kerf

Table No: 5

2.3.1.10 MATERIAL DIMENSIONS

Length (mm)	Qty	Name
200	1	leadscrew length
1054	2	belt length along x
1063	2	belt length along y
4234	**total length**	belts (all 4)

Table No: 6

2.3.1.11 TOTAL MACHINE FOOTPRINT

Length (mm)	Name
1072	x
1081	y
540	z
680	clearance to remove z axis

Table no: 7

2.3.2 DESCRIPTION OF COMPONENTS

2.3.2.1 PRINTED PART LIST:

QTY	Part Name	Infill	Color
	Corner		
2	Corner Bottom	45%+	B
2	Corner Bottom Mirrored	45%+	B
2	Corner Top	45%+	A
2	Corner Top Mirrored	45%+	A
2	Lower Belt	45%+	A
2	Lower Belt Mirrored	45%+	A
2	Upper Belt	45%+	B
2	Upper Belt Mirrored	45%+	B
4	Corner Leg Lock	45%+	B
4	Feet	45%+	A
2	Wire Darryl	45%+	A
4	Stop Block (Dual only)	45%+	A
	Trucks		
2	Truck	45%+	A
2	Truck Mirrored	45%+	A
4	Truck Clamp	45%+	B

QTY	Part Name	Infill	Color
	Center Assembly		
2	Core Z Clamp 1	45%+	A
2	Core Z Clamp 2	45%+	A
3	Core Clamp	45%+	A
1	Core ClampY	45%+	A
1	Core	70%	B

Table no: 8

2.3.2.2 HARDWARE:

Hardware	Type	Alternative	QTY
5/16x1.5"	Bolt	M8x40mm	44
5/16 locknut		M8 locknut	44
M5x30mm	Screw		60
M5 locknut			60
M3x10mm	Screw		22
M2.5x12mm	Screw	#3x½"	8

Table no: 9

2.3.2.3 COMPONENTS:

Part	QTY
Nema 17 50OZ/in+	5
* Belt GT2 10mm	4
Pulley 16t gt2 10mm (5mm bore)	4
Idler 20t gt2 10mm (5mm bore)	8
Power Supply	1
Wiring kit	1
Bearings	45
leadscrew/nut	1
5mm to 8mm Coupler	1
Lube	1
Control Board	1

Table no 3

- **Nema 17 Stepper Motor:**

  - **Rated Voltage** **:** 12V DC.
  - **Current** : 1.2A at 4V**.**
  - **Step Angle** : 1.8 deg.
  - **No. of Phases** : 4.
  - **Motor Length** : 1.54 inches**.**
  - **4-wire, 8 inch lead.**
  - **200 steps per revolution** : 1.8 degrees.
  - **Operating Temperature** : -10 to 40 °C.
  - **Unipolar Holding Torque** : 22.2 Oz-in.

- **GT2 Belt 10mm:**

  - **Width –** 10mm.
  - **Tooth Pitch –** 2mm.
  - **Material –** Nylon Cover / Fibreglass Reinforced / Neoprene.

- **Pulley GT2 :**

  - **Bore Diameter (B) (mm)** **-**5mm
  - **Outer Diameter (OD) (mm)** **-**16mm
  - **Pitch (GT2)** **-**2mm
  - **Belt width** **-**10mm
  - **No. of teeth** **-**16 teeth
  - **Length (mm)** **-**16mm
  - **Weight (gm)** **-**12gm

- **POWER SUPPLY:**

  - **DC Voltage** : 12V
  - **Rated Current** **:** 5A
  - **Current Range** **:** 0-5A
  - **Rated Power** **:** 60W
  - **Ripple & Noise (maximum)** : 150mV
  - **Voltage Adjustable Range** **:** 10-13.2V
  - **Voltage Tolerance** **:** ±2.0%
  - **Line Regulation** **:** ±0.5%
  - **Load Regulation** **:** ±0.5%

- **Arduion UNO:**

  - **Microcontroller** : ATmega328P
  - **Operating Voltage** : 5V
  - **Input Voltage (recommended)** : 7-12V
  - **Output Voltage (limit)** : 6-20V
  - **Digital I/O Pins** : 14
  - **PWM Digital I/O Pins** : 6
  - **Analog Input Pins** : 6
  - **DC Current per I/O Pin** : 20 mA
  - **DC current for 3.3V Pin** : 50 mA
  - **Flash Memory** : 32 KB
  - **SRAM** : 2 KB
  - **EEPROM** : 1 KB
  - **Clock Speed** : 16 MHz
  - **LED_BUILTIN** : 13
  - **Length** : 68.6 mm
  - **Width** : 58.4 mm
  - **Weight** : 25 g

- **CNC SHIELD:**

  - **Motor Voltage** : 8 V to 35 V.
  - **Logic Circuits Voltage** : 3 V to 5.5 V.

- **Current** : 2 A (MAX).
- **Five step resolutions** : full, 1/2, 1/4, 1/8 and 1/16.
- **Protection** : under-voltage, over-current and over-temperature.

- **A4988 Stepper Motor Driver:**

  - **Minimum operating voltage** -8 V
  - **Maximum operating voltage** -35 V
  - **Continuous current per phase** -1 A
  - **Maximum current per phase** -2 A
  - **Minimum logic voltage** -3 V
  - **Maximum logic voltage** -5.5 V
  - **Microstep resolutions** -1, 1/2, 1/4, 1/8, 1/16
  - **Size** -0.6″ × 0.8″
  - **Weight** -1.3 g

- **LASER MODULE**

  - **Wave length** : 450nm.
  - **Power** : 5500MW.
  - **Electric current** : <5A.
  - **Input voltage** : DC AC 12V.
  - **Working temperature** : -10 ~+40.
  - **Size** : 33*33*75mm.

CHAPTER 3

3.1 APPROACH AND METHODOLOGY

3.1.1 TECHNOLOGY USED:

- **Laser Technology:** CNC laser engravers use laser technology to create the engraving, which emits a highly focused beam of light that can be precisely controlled to remove material from the surface and create the desired marks or patterns. Common types of lasers used in engraving machines include CO_2 lasers, fibre lasers, and diode lasers.
- **CNC (Computer Numerical Control):** CNC technology is built into laser engraver equipment. It entails accurately controlling the movement of the laser head or the work piece using computer software and control systems. The CNC system analyses digital drawings or instructions and converts them into synchronised laser motions, allowing for precision engraving.
- **Motion Control:** Motion control systems are used by CNC laser engravers to control the movement of the laser head or the work piece. Stepper or servo motors are frequently used in conjunction with linear guides or belts to move the laser or work piece along several axes (normally X, Y, and occasionally Z). The motion control system guarantees that the laser is accurately positioned and moved for precision engraving.
- **Cooling Systems:** Because laser engraving creates heat, cooling systems are required to keep the laser components at the appropriate temperature. These systems frequently use water or air cooling to minimise overheating and preserve the laser's lifespan and effectiveness.

- **Optics and Beam Delivery:** Laser engraver machines control and guide the laser beam using a set of lenses. Mirrors and lenses are commonly used in these optics to concentrate and direct the laser beam onto the work piece. Beam delivery systems, such as articulated arms or gantry systems, allow the laser beam to move around the work piece to engrave different locations.
- **Control Software:** CNC laser engraving devices require specialised control software to function. Users may use this programme to import or design digital files, select engraving parameters, move the laser or work piece, and monitor the engraving operation. It also allows you to customise the laser strength, speed, and other variables to obtain the desired results.

Based on the problem definition of developing a "CNC LASER ENGRAVER MACHINE", the following methodology can be considered:

1. **Complex Design**: Developing a CNC laser engraver machine requires expertise in mechanical engineering, electrical engineering, and computer programming. The design of the machine needs to be highly precise and complex to ensure its accuracy and reliability.

2. **High Development Cost**: Developing a CNC laser engraver machine can be expensive due to the specialized components and materials required, as well as the advanced technology needed to control the machine. This can make it challenging to keep the project within budget.

3. **Customization and Flexibility**: CNC laser engraver machines are often customized to meet specific manufacturing requirements, which can make it difficult to develop a machine that is flexible enough to meet the needs of multiple customers. This may require additional development time and cost.

4. **Testing and Certification**: CNC laser engraver machines are required to meet various safety standards and regulations, which can add to the development time and cost. The machine must also be thoroughly tested to ensure its reliability and accuracy.

5. **Competition**: The market for CNC laser engraver machines is highly competitive, and there are many established manufacturers offering similar machines. This can make it challenging to differentiate the machine and establish a foothold in the market.

CNC laser engraver machines have a wide range of use cases across various industries. Some common applications include:

- **Personalized and Customized Items**: CNC laser engravers may be used to manufacture personalised and customised goods such as jewellery, presents, trophies, and awards. They enable detailed and precise engraving on a variety of materials such as wood, acrylic, metal, and leather.
- **Signage and Branding**: CNC laser engravers are extensively used in the signage sector to make high-quality signs, nameplates, and logos. They can engrave on materials such as acrylic, wood, and metal, resulting in professional and long-lasting signage solutions.
- **Industrial Marking and Serializing**: CNC laser engravers are used for industrial marking and serialising, such as engraving serial numbers, barcodes, QR codes, and identifying tags on products and components. This aids traceability, inventory management, and anti-counterfeiting efforts.
- **Art and Decorative Objects:** Artists and designers use CNC laser engraving equipment to produce elaborate patterns, artwork, and ornamental objects on a variety of materials. It enables accurate details as well as the replication of complicated patterns.

- **Architectural Models and Prototypes:** Architectural models and prototypes are created using CNC laser engravers. They can engrave precise shapes and textures onto materials such as wood and foam, making it easier to visualise architectural plans and speed up the prototype process.
- **Woodworking and Furniture Making:** CNC laser engravers are frequently used in the woodworking sector to engrave elaborate designs, patterns, and trademarks into hardwood furniture, cabinets, and ornamental objects.
- **Electronics and PCB Manufacturing:** Electronics and PCB Manufacturing: CNC laser engravers are used in the electronics sector to create accurate tracks, labels, and markings on printed circuit boards (PCBs). They also make it easier to manufacture specialised electrical components.

3..2 PROGRAMMING:

Grbl is a no-compromise, high performance, low cost alternative to parallel-port-based motion control for CNC milling. It will run on a vanilla Arduino (Duemillanove/Uno) as long as it sports an Atmega 328.

The controller is written in highly optimized C utilizing every clever feature of the AVR-chips to achieve precise timing and asynchronous operation. It is able to maintain up to 30 kHz of stable, jitter free control pulses.

It accepts standards-compliant g-code and has been tested with the output of several CAM tools with no problems. Arcs, circles and helical motion are fully supported, as well as, all other primary g-code commands. Macro functions, variables, and most canned cycles are not supported, but we think GUIs can do a much better job at translating them into straight g-code anyhow.

Figures no: 13

3..3 FABRICATION:

- **Legs**

Figures no: 14

- The leg lengths come from the cut calculator.
- Do whatever it takes to get the legs as equal in length as possible. This will really helpin levelling and keeping it level.
- In this picture, the one on the far left is too short, so it could either be shimmed duringassembly or sand/grind the other legs.
- A little extra time here pays off for the life of the machine.

Tip:

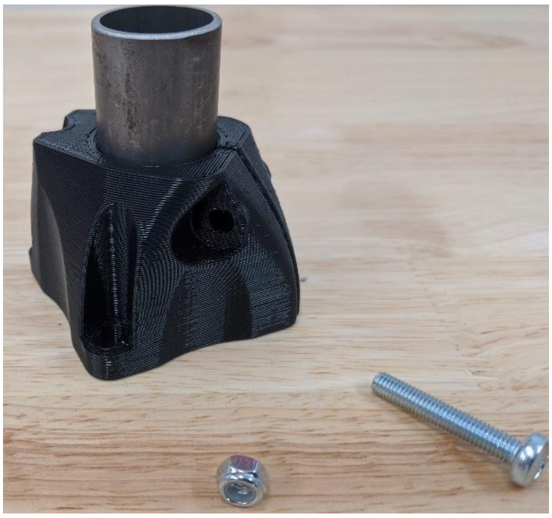
Sand, grind, or file the inside and outside edges of all the cuts on the rails and legs. Thisprevents cut wires and eases assembly.

- **Tools:**

Figures no: 15

- 8mm Socket or nut driver.
- Phillips #2 Screwdriver.

- **Legs and Feet**



Figures no: 16

- You will need the legs, printed feet, M5x30 screws and locknuts.

Tip:

It is best to pre-thread all the locknuts before assembly. This will loosen them and preventmangling of the printed parts.

Figures no: 17

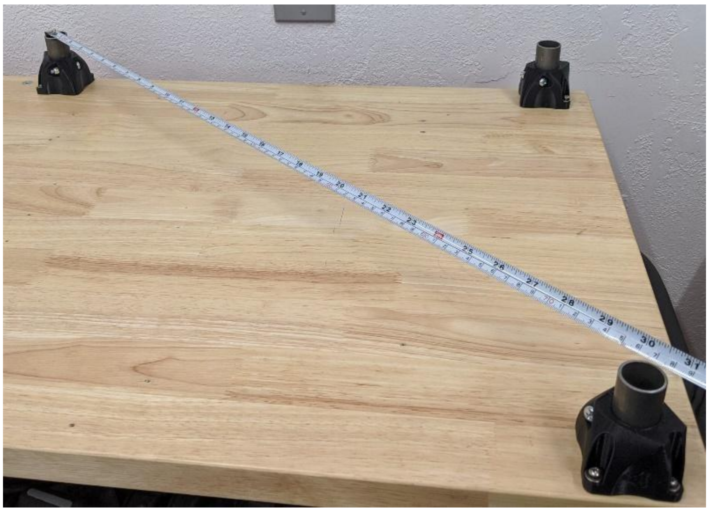
- Insert the leg and make sure it is seated all the way to the table.
- The feet have a small side for the head of the screw and a larger opening for the nutand socket.
- Snug this screw. Snug means just a ¼ turn past the nut making contact. This does notneed to be super tight.
- There is a 2+mm gap here, do not try to close the gap.

- **BASE LAYOUT:**

Figures no: 18

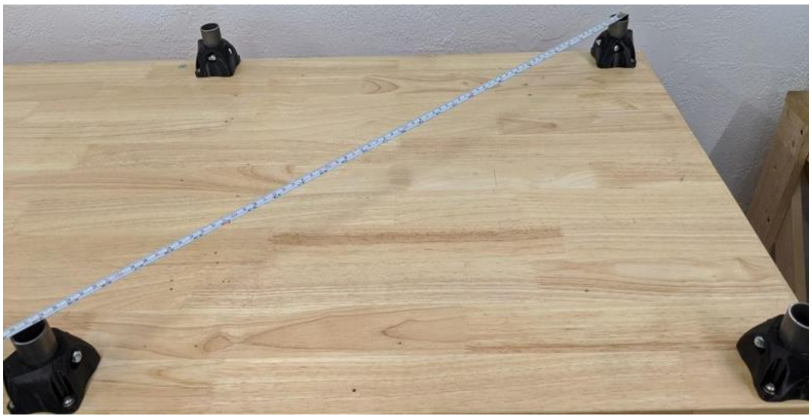
- Use whatever screws are appropriate for your table, these are not included in the V1 Kits.

- Layout the foot assemblies on your table according to the dimensions from the cut calculator.
- Notice all the clamping screws are oriented towards the centre of the build area.
- Screw down the two assemblies along the front edge of the table. These will be your fixed references.
- Measure as carefully and accurately as you can across the leg tops. Your machine can only be as accurate as you build it.
- Validate that the front length exactly matches the back length, and that the left and right lengths match.
- Measure diagonally across the leg tops.



Figures no: 19

- Compare this measurement to the next one.



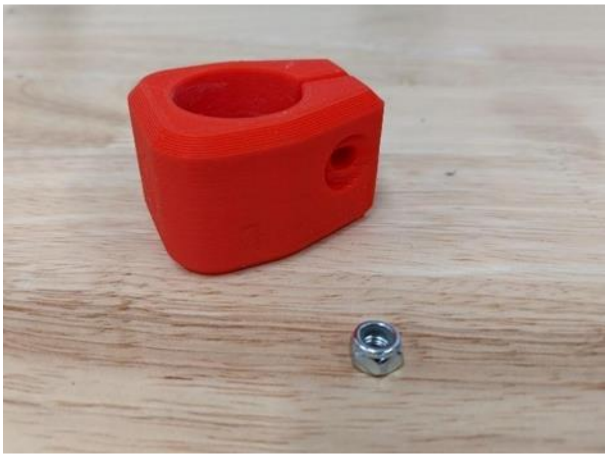
Figures no: 20

- This next diagonal should match the previous one. Squaring this way can be extremely accurate because any error in the leg locations is multiplied in this measurement.
- Adjust both rear legs until you are satisfied with the precision.
- Screw in one foot at a time, minor adjustments are possible.
- Triple check the diagonals - they need to be as close as you can get them.

Tip:

Marking and pre-drilling the foot hold downs in the centre of the screw slots makes it much easier to get a square base. It allows a tiny bit of room for adjustment.

- **LEG LOCKS:**



Figures no: 21

- You will need the printed Corner Leg Locks and M5 locknuts.
- It is best to pre-thread all the locknuts before assembly. This will loosen them and prevent mangling of the printed parts.

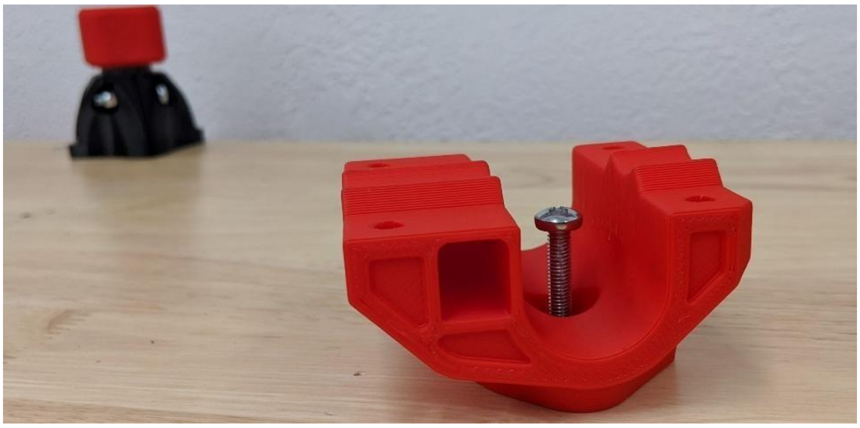
Figures no: 22

- Push the nut into the Corner Leg Lock and fully seat it. If it does not stay on its own you can use a drop of glue
- If the nut does not stay on its own, you can attach the Corner Bottom very loosely with its bolt.

Figures no: 23

  - Fully seat the Corner Leg Locks onto the legs.
  - Orient to match the Feet below.
  - Do not lock/clamp them.

- **CORNER BOTTOMS**



Figures no: 24

  - You will need the printed Corner Bottoms and Corner Bottom Mirrored and M5x30 screws

Figures no: 25

- The Corner Bottoms go on the Leg Locks as shown. Pay attention to how the belt slider slots are oriented. They point towards each other on the outside of the build area.
- This picture is taken with the 0, 0 (home) corner in the bottom nearest left. The corners should be as shown.

Figures no: 26

- Get the screws started to hold the Leg Locks and Corner Bottoms together.
- Leave them loose, just tight enough to make the screw head clear the rail.
- Temporarily adding the Y rails will align the corners.

Figures no: 27

- **LOCK THE LEGS**

Figures no: 28

- You will need four sets of the M5 screws and locknuts.
- It is best to pre-thread all the locknuts before assembly. This will loosen them and prevent mangling of the printed parts.

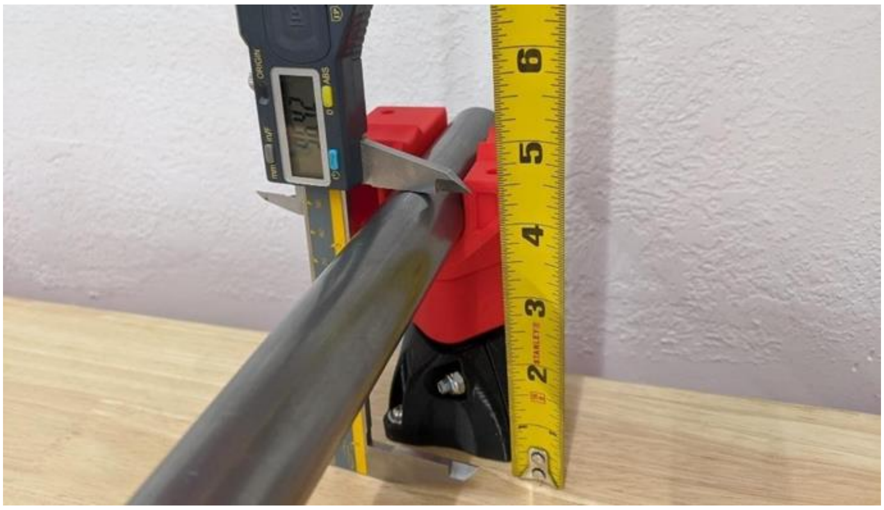
Figures no: 29

- Snug up all four Leg Locks.
- Snug is just that, ¼ turn past the screw and nut seating into the part. Just tight enough to make the corner have some resistance if you try to take it off the leg.
- All clamping parts have a gap. Do not clamp it hard enough to close the gap!
- Now remove the rails and snug up the screw in the bottom of the corner pieces.

Figures no: 30

Tip: This screw is tightened after the Leg Locks so the Corner Bottom will correctly and fully seat on the angled joint between the two parts. If you tighten the bottom first, you might have a hard time locking the leg into place or the Corner Bottom could come loose.

- **LEVELLING:**



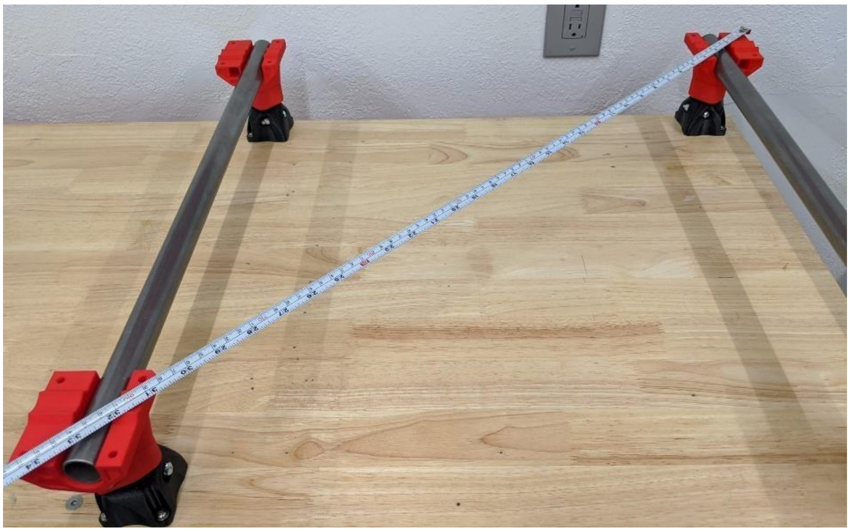
Figures no: 31

  - Put the Y rails back in place.
  - Use whatever means you have at your disposal to assure all four corners are at the same level.
  - If you were careful with your leg lengths this should be level already.
  - Loosening the screws and giving the corners a light tap can seat them a tiny bit farther if needed.

- **FINAL CHECK:**

Figures no: 32

- Best practice is to check your work at every step.
- Measure the diagonal measurement to the best of your abilities.



Figures no: 33

- This diagonal should match. You should aim for no more than 1 mm deviation.
- If there is an issue, loosen the screws into the table and nudge a leg assembly.

- **ASSEMBLING THE TRUCKS**
  - **BEARINGS**

Figures no: 34

- You will need 5 sets of 5/16 x 1 ½" (M8 x 40mm) bolts and locknuts.
- Two ½" sockets and ratchets (or 13mm for M8's).

- 5 bearings.
- Printed Truck and Truck Mirrored parts.

- **TENSION BOLTS**

Figures no: 34

- Insert the bearings in the Truck and fit the bolts.
- Just seat the locknuts, they will get adjusted later. Leave them loose.

Note:

Pay attention to the direction of the bolt heads on these two tension bolts.

- **STATIONARY BEARINGS**
  - Fit the last 3 bearings and bolt sets.
  - Notice the bolt orientation, nuts towards the rails.
  - Snug these up. About 7in/lbs.

Figures no: 35

Caution:

Using ratchets makes over tightening the bolts very easy. There are small 1mm built in spacers in the printed parts, if you crush these you are about 3x too tight. Loose is better thantight. If you work on your own car, it is way less that any bolt you have ever tightened.

- **SET THE TRUCK TENSION**

Figures no: 36

- Insert a rail into the Truck.
- You are looking for all bearings to make contact and just a hint of tension. If you tip the rail at a 45 degree angle the Trucks should slowly move. Too tight and theywill stay, too loose and they will drop right off.
- Tension both bolts evenly - for a new print this should actually just be good with no extra tension needed. If this is the case, just snug up the tension bolts until the heads and nuts are touching the plastic. Holding not adding to the current tension.

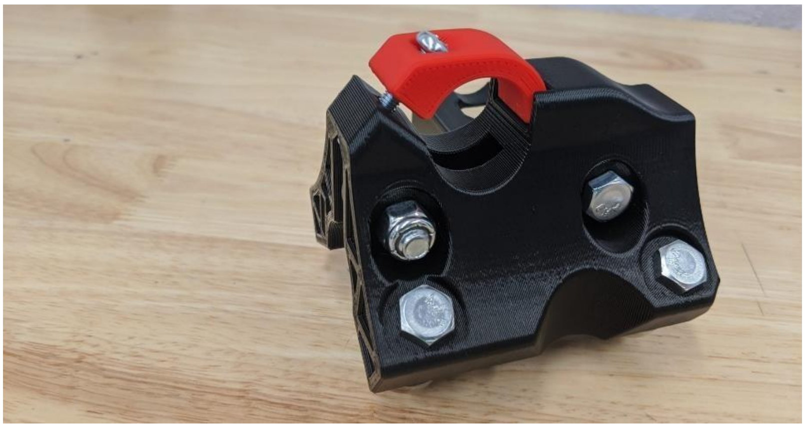
Caution:

If you have to really torque down on these bolts you could have the wrong sized rails or parts.The same thing applies if they are too tight, although you can let them sit overnight with no tension on the bolts to see if they loosen up a bit.

- **TRUCK CLAMP**

Figures no: 37

  - You will need the printed Truck Clamps.
  - M5 x 30mm screw and locknut.
  - #2 screwdriver.



Figures no: 38

- Leave these loose, just engaging the nut locks. The rail will be added later.

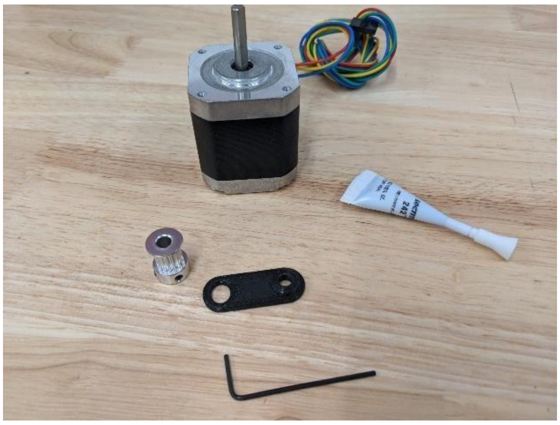
Tip:

For all the screws with captured nuts it is best to pre-thread them to loosen them up a bit toprevent ruining the printed parts.

- Tuck in the idlers into the pocket in the Trucks.
- Tighten the screws just tight enough to seat the nut all the way then back off a ½ turn.

- Ensure the idlers are free spinning.

- **PULLEYS**

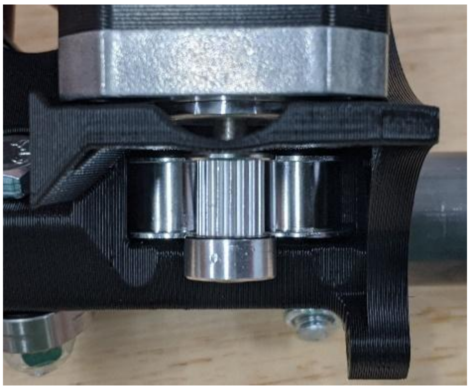


Figures no: 39

- Steppers
- Pulleys
- Wire Darryl / pulley spacer
- (optional) Loctite

Figures no: 40

- You can use the Wire Darryl to set the pulley height or just do it by eye in the nextstep.



Figures no: 41

- Make sure the pulley is aligned with the idlers.
- (optional) Add Loctite to the grub screws, even a bit on the stepper shaft can help.
- Tighten the grub screw on the stepper shaft flat first.
- Tighten the second grub screw.

Tip:

Loose pulleys are probably the number one issue we see on both the CNC Laser engraver machine builds. They cause all sort of erratic behavior and false “skipped steps”. Take your time and do this correctly. Notice it does not say tighten the heck out of them. Get them on in the right order,directly on the flat, and Loctite is your friend.

- **STEPPERS**

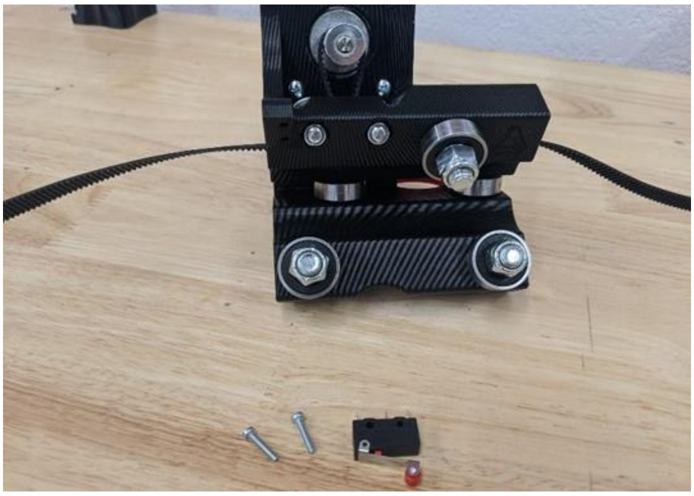
Figures no: 42

- M3 x 10mm screws.
- #1 screwdriver.
- Truck assemblies and stepper assemblies.

Figures no: 43

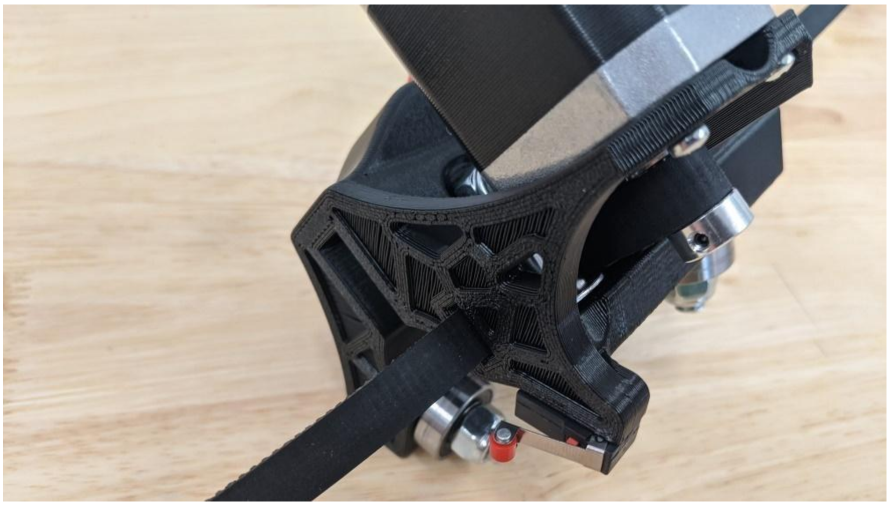
- Snug down the screws evenly.
- Double-check the pulley and idlers are still aligned.

- **OPTIONAL END STOPS**



Figures no: 44

- M2.5 x 12 screws (or #3 x ½” wood screws).
- #1 screwdriver.
- Limit switches.
- Orient the end stop so the trigger is nearest the belt and rail.



Figures no: 45

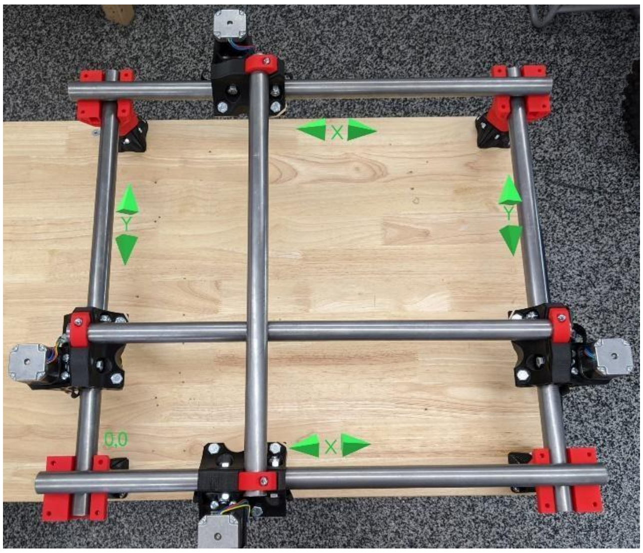
- **CORNER TOPS AND RAILS:**

  - **TOPS**

Figures no: 46

- You will need the printed Corner Top and Corner Top Mirrored parts.
- M5 screws and nuts.
- All 4 X and Y rails.
- Truck assemblies from previous step.
- #2 Screwdriver.

- **LAYOUT**



Figures no: 47

- Use this picture to orient the Trucks and sort out your X and Y rails
- You do not need the gantry rails at this point, they are just in this picture to clarifythe position of the Trucks.

Figures no: 48

- Add the tops.

- Pay close attention to the belt slides: they should be on the opposite rails of theCorner Bottom parts.
- Add the screws and nuts and lightly seat them all.

Caution;

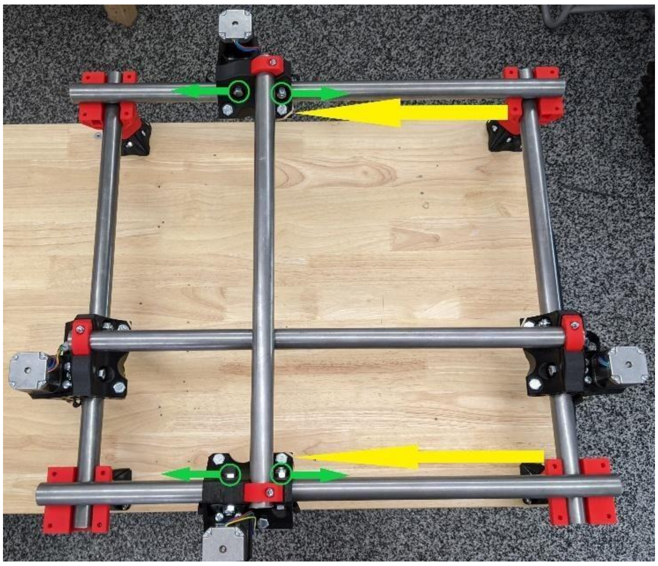
With all the captured nuts you should pre-thread the screws at least once to loosen thelocknuts up a bit to not destroy the printed parts.

Figures no: 49

- Notice the even gap in the corner pieces. They all should have this.
- Slightly snug down the screws evenly, leaving an even gap on all sides.
- Snug means the rails will still be able to rotate if you try. You should feel someresistance though.
- Do not over tighten these screws

- **TRUCK SQUARING**
  - New to the Primo version, before moving on it is best to square your Trucks.
  - Clamp in your gantry rails.
  - Move the axis around to settle it.
  - Measure from the Trucks on the same axis to either corner (yellow arrow).

- Now you can move the Trucks by very slightly tightening the tension bolts.



Figures no: 50

- Start small: try 1/16th of a turn (green arrows).
- Too tight and it will stop moving.
- Loosening will relax it opposite the green arrows.
- Check both axes, again looking for 1mm or under.
- Remove the gantry rails.

CHAPTER 4

4.1 TESTS AND VALIDATION

4.1.1 TEST PLAN:

- **Introduction**
  - **Objective:** This test plan's objective is to explain the testing strategy for the CNC Laser Engraver Machine project.
  - **Scope:** The test plan will include the CNC Laser Engraver Machine's functional and non-functional testing operations.

- **Objectives**
  - To confirm that the CNC Laser Engraver Machine fits the criteria.
  - To discover and disclose any flaws or problems.
  - To validate the machine's performance, dependability, and safety.
  - To guarantee that the CNC Laser Engraver Machine works well with various materials and designs.

4.1.2 TEST APPROACH

- **Unit Testing:** Individual components of the CNC Laser Engraver Machine will be separately tested to ensure their operation.
- **Integration Testing:** The integration of various components will be tested to verify that they perform flawlessly together.
- **System Testing:** The entire CNC Laser Engraver Machine will be checked to ensure its general operation.
- **Performance Testing:** To confirm that the machine satisfies the specified standards, its performance will be evaluated under varied workloads.
- **Usability Testing:** The machine's usability will be evaluated from the perspective of an end-user to determine its user-friendliness and simplicity of operation.
- **Safety Testing:** The machine's safety features will be evaluated to verify they work as intended and meet safety regulations.
- **Compatibility Testing:** The machine will be tested with various materials and designs to guarantee that it operates well in a variety of circumstances.
- **Regression Testing:** Following each modification or update, regression testing will be performed to ensure that new changes do not impair current functionality.

4.1.3 FEATURES TESTED

- **Power Output:**

  The laser engraver uses a 5.5W laser module to determine the intensity and depth of the engraving.

- **Engraving Area:**

  The machine has a working area of 700mm/700mm.

- **Frame and Construction:**

  To maintain stability throughout the engraving process, the machine have a solid and durable frame. To offer stiffness, we have used stainless steel and 3D printed parts.

- **Controller Software:**

  The machine is compatible with laserGRBL software, which serves as the engraving process's controller. We use the programme to input designs, control the laser's movement, change power settings, and manage other factors.

- **Communication:**

  The laser engraver may feature many communication options, such as USB which allow you to connect it to a computer.

- **Safety measures:**

  Laser engravers are often equipped with safety measures to keep users safe. These incorporate laser radiation-containment enclosures, emergency stop buttons, and safety interlocks to avoid unintentional exposure to the laser beam.

- **Software Features:**

  LaserGRBL software provide functionality for design modification, vector tracing, picture manipulation, and laser parameter control. It also include settings for engraving speed, power, and resolution.

- **Cooling System:**

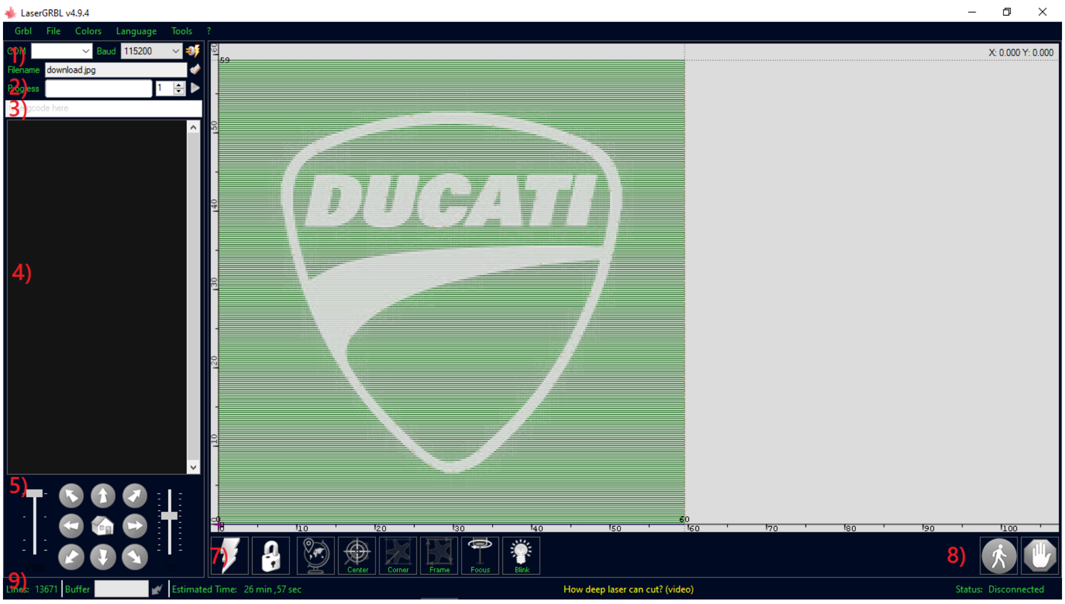
  A 5.5W laser engraver normally require a cooling system, such as an air mechanism, to disperse the heat created during the engraving process and prevent laser module damage.

4.1.4 FINDINGS

We found out that the machine we built is a very accurate and a tough competition to the competitor company like creality, wol3D and man6y more company's.

4.1.5 INFERENCE

We have used an open source software called "LaserGRBL".



1. Connection control: here you can select serial port and proper baud rate for connection, according to grbl firmware configuration.
2. File control: this show loaded filename and engraving process progress. The green "Play" button will start program execution.
3. Manual commands: you can type any G-Code line here and press "enter". Commands will be enquired to command queue.
4. Command log and command return codes: show enquired commands and their execution status and errors.

5. Jogging control: allow manual positioning of the laser. The left vertical slider control movement speed, right slider control step size.
6. Engraving preview: this area show final work preview. During engraving a small blue cross will show current laser position at runtime.
7. Grbl reset/homing/unlock: this buttons submit soft-reset, homing and unlock command to grbl board. On the right of unlock button you can add some user defined buttons.
8. Feed hold and resume: this buttons can suspend and resume program execution sending Feed Hold or Resume command to grbl board.
9. Line count and time projection: LaserGRBL could estimate program execution time based on actual speed and job progress.

CHAPTER 5

5.1 BUSSINESS ASPECTS

- **VERSATILITY:**

  CNC laser engravers are very adaptable equipment that can engrave and mark a broad range of materials, including wood, metal, plastic, glass, leather, and others. This adaptability enables firms to provide a varied range of products and services to meet the demands of their customers.

- **PRECISION AND QUALITY:**

  Laser engraving is extremely precise and creates high-quality results. CNC laser engravers' precision and exquisite detail make them excellent for a variety of applications, including bespoke signs, personalised items, branding, and industrial marking.

- **TIME AND COST EFFICIENCY:**

  CNC laser engravers can accomplish engraving operations fast and efficiently, minimising manufacturing time when compared to older techniques. CNC control systems minimise the need for manual labour, lowering operational costs and eliminating mistakes.

- **CUSTOMIZATION AND PERSONALIZATION:**

  Businesses may simply provide customised and personalised items with CNC laser engraving. This expertise enables businesses to target specialised markets and offer one-of-a-kind solutions such as personalised gifts, promotional goods, and bespoke designs.

- **SCALABILITY:**

  CNC laser engraving machines are capable of handling both small and big production runs. This scalability enables organisations to grow and cater to a variety of order sizes, from individual orders to mass production.

- **INTEGRATION WITH DIGITAL WORKFLOWS:**

  CNC laser engravers may function in conjunction with digital design tools and workflows. This connection streamlines the production process and allows for fast iterations and revisions by facilitating the transmission of digital drawings to the machine.

- **REDUCED WASTE AND ENVIRONMENTAL IMPACT:**

  Laser engraving is a non-contact procedure that reduces material waste and has a low environmental impact. Laser technology's accuracy assures fewer defects, decreasing the requirement for rework or discarded materials. Furthermore, because laser engraving does not utilise chemicals or inks, it is an ecologically beneficial solution.

- **DIVERSIFICATION AND MARKET OPPORTUNITIES:**

  Entrepreneurs may tap into a variety of sectors and industries by introducing CNC laser engravers into their operations. These machines are used in areas such as signs, jewellery, electronics, automotive, aerospace, architecture, and many others, bringing up new revenue sources.

- **MAINTENANCE AND TRAINING:**

  It is critical to address the maintenance and training requirements connected with CNC laser engravers. Regular maintenance and calibration guarantee maximum machine performance, while thorough operator training programmes maximise productivity and reduce mistakes.

- **COMPETITIVE ADVANTAGE:**

  Using CNC laser engraving technology may provide businesses with a competitive advantage by delivering unique goods, shorter turnaround times, and greater quality when compared to rivals that use traditional engraving methods.

5.2 FINANCIAL CONSIDERATIONS

- **INITIAL INVESTMENT:**

  The cost of purchasing the CNC laser engraving machine is a crucial aspect. This covers the machine's purchase price as well as any additional equipment or accessories required for operation. Investigate several providers and models to get a sense of the price range.

- **OPERATING COSTS:**

  Determine the continuous costs of running the CNC laser engraver. This covers fees for energy, maintenance and repair, replacement components, and consumables including laser tubes, lenses, and cooling fluids. These expenses must be considered in order to estimate the long-term financial effects.

- **LABOR COSTS:**

  Consider the labour costs associated with operating the equipment. Calculate the associated wages or salaries if you intend to engage operators. Furthermore, training fees may be spent to guarantee that the operators are capable of running the equipment safely and effectively.

- **Costs of Facility:**

  Determine the space required for the CNC laser engraver machine. Check that your facility has enough space and factor in any additional costs involved with setting up or changing the workplace. This might include ventilation systems, safety precautions, and ergonomic concerns.

- **Maintenance and Training:**

  It is critical to address the maintenance and training requirements connected with CNC laser engravers. Regular maintenance and calibration guarantee maximum machine performance, while thorough operator training programmes maximise productivity and reduce mistakes.

5.3 CONCLUSIONS

5.3.1 DESCRIBE STATE OF COMPLETION OF CAPSTONE PROJECT.

Finally, the CNC laser engraver machine is a highly efficient and adaptable instrument that provides significant advantages in a variety of sectors. Throughout this research, we have investigated the machine's capabilities, benefits, and prospective uses.

To begin with, the CNC laser engraving machine has excellent precision and accuracy, allowing users to produce elaborate and precise drawings on a variety of materials. Because it is computer-controlled, it produces consistent outcomes while reducing human error and enhancing productivity.

Another important feature is the machine's adaptability. It can engrave on a variety of materials such as wood, acrylic, glass, metal, and others. This adaptability enables it to serve a wide range of businesses, including manufacturing, arts & crafts, signs, jewellery creation, and personalised gift manufacture.

Furthermore, compared to traditional engraving processes, the CNC laser engraver machine has significant benefits. It removes the need for actual templates or stencils, saving time and money during setup. It also allows for quick prototyping, which makes it perfect for small-scale production and customisation.

Furthermore, the machine's software compatibility allows for simple design generation and customization. Users can use specialist programmes to develop their own designs or import designs from popular graphic design applications. This user-friendly interface allows both pros and amateurs to explore their creativity and generate high-quality results.

The CNC laser engraving machine also has safety measures to safeguard users and reduce the risks connected with laser use. These safety precautions, which include emergency stop buttons, protective enclosures, and beam shields, help to provide a safe working environment.

Overall, the CNC laser engraving machine is a sophisticated instrument that enables users to realise their visions with precision, efficiency, and variety. It has a wide range of possible uses, and it is predicted to continue revolutionising sectors by improving manufacturing processes, increasing creativity, and boosting innovation.

5.3.2 FUTURE WORK

By using this machine we can start our start up with help of using platform like Instagram in which we could take orders and with the help of the "CNC Laser engraver machine" we can make personalised gifts as per requirements also we can use in the industry for small work holds.